

Instructor Guide & Student Lab: GitHub Projects Automation

Prerequisites

- Each student (or pair) needs a **GitHub account**.
- A sample public/private repository created (e.g., github-workflow-demo).
- A **GitHub Project (v2)** board created and linked to the repository.

Step 1: Set Up the Project Board Columns & Workflows

1. Navigate to your repository on GitHub.
2. Click the **Projects** tab \rightarrow Click **New project** \rightarrow Choose **Board** template \rightarrow Name it **Sprint Board**.
3. Configure the board columns:
 - **Todo**
 - **In Progress**
 - **In Review**
 - **Done**
4. Open the Project Workflows menu:
 - Click the ... (three dots menu) at the top right of the project board \rightarrow Select **Workflows**.
5. Enable the following workflows:
 - ☒ **Auto-add to project** (set filter to your repo)
 - ☒ **Item added to project** (set default status to Todo)
 - ☒ **Auto-add sub-issues to project**
 - ☒ **Pull request linked to issue** (set status to In Progress)
 - ☒ **Pull request merged** (set status to Done)
 - ☒ **Auto-close issue**

Step 2: Demonstrate "Auto-Add" & Default Status

1. Go back to the **Repository** page.
2. Click **Issues** \rightarrow Click **New issue**.
3. Create an issue:
 - **Title:** Issue #1: Fix header responsiveness on mobile
 - **Submit** the issue.
4. **Student Check:** Switch to the **Projects** tab.
 - *Observation:* Notice how Issue #1 automatically appeared on the board inside the **Todo** column without any manual dragging!

Step 3: Demonstrate "Sub-Issues Automation"

1. Open Issue #1 in the repository.
2. Scroll to the **Create sub-issue** section (or task list).
3. Add a new sub-issue:

- **Title:** Sub-issue: Update CSS breakpoints
- 4. **Student Check:** Switch back to the **Projects** board.
 - *Observation:* The sub-issue automatically appears on the project board alongside the parent issue.

Step 4: Demonstrate "PR Linking & Status Change"

1. Open your terminal or GitHub web interface and create a new Pull Request.
2. Title the PR: Fix CSS media queries.
3. In the **PR description**, type keyword linking syntax: Fixes #2 (*Replace #2 with the sub-issue number generated above*)
4. Submit the Pull Request.
5. **Student Check:** Look at the Project Board.
 - *Observation:* The sub-issue card automatically moved from **Todo** → **In Progress** as soon as the PR was linked.

Step 5: Demonstrate "PR Merging & Auto-Done"

1. Open the created Pull Request.
2. Click **Merge pull request** → **Confirm merge**.
3. **Student Check:** Switch to the Project Board.
 - *Observation:* Both the **Pull Request** and the linked **Sub-issue** have automatically moved to the **Done** column.

Step 6: Demonstrate "Board-Driven Auto-Closing"

1. Find the original parent issue (Issue #1) on the Project Board under **Todo**.
2. Click and drag Issue #1 into the **Done** column.
3. **Student Check:** Open Issue #1 on the repository Issues page.
 - *Observation:* GitHub automatically changed the state of Issue #1 from Open to **Closed** because it was dropped into Done.

Key Takeaways for Students

- **Zero Overhead:** Developers spend less time updating project boards and more time coding.
- **Traceability:** Pull Requests explicitly track which issues they solve using standard keywords (Fixes #ID, Closes #ID).
- **Single Source of Truth:** Board status accurately reflects real git activity in real-time.